

ALNS — Adaptive Large Neighbourhood Search (Ropke & Pisinger 2006)

Destroy q nodes, repair with an insertion heuristic, accept via a pluggable criterion (default simulated annealing), and learn operator weights per segment from scores $\sigma_1/\sigma_2/\sigma_3$. Objective: maximise profit = $R \cdot \Sigma \text{ waste} - C \cdot \Sigma \text{ distance}$ under capacity and mandatory coverage.

